American Computer Science League

2021-2022 ● Contest 3: Short Problems Solutions ● Classroom Division

1. Boolean Algebra

$$\overline{A}\,(AB + \overline{B}) = A\,\overline{A}\,B + \overline{A}\,\overline{B}$$
$$= 0 + \overline{A}\,\overline{B}$$
$$= \overline{A}\,\overline{B}$$

To be true both $\overline{A}$ & $\overline{B}$ must be 0.
So (0, 0) is the only ordered pair that satisfies the condition.

A. (0, 0)

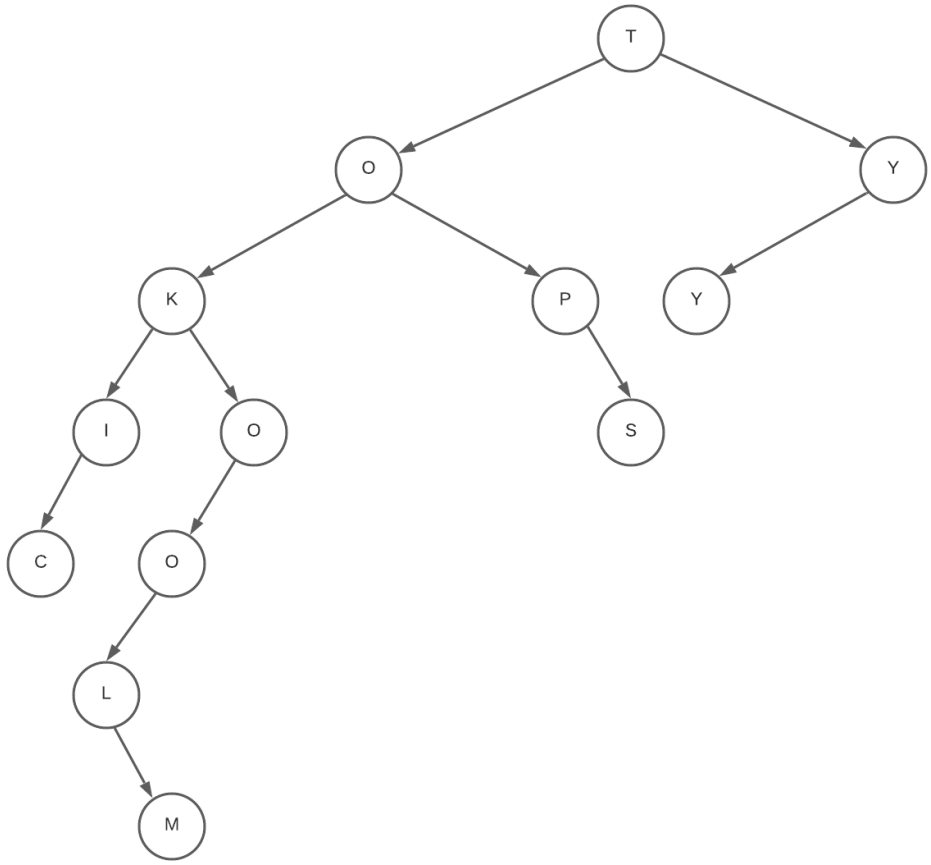
2. Boolean Algebra

$$A\,(\overline{B} + \overline{C}) + \overline{A}\,B + \overline{B\,C} = A\,\overline{B} + A\,\overline{C} + \overline{A}\,B + \overline{B} + \overline{C}$$
$$= (A\,\overline{B} + \overline{B}) + (A\,\overline{C} + \overline{C}) + \overline{A}\,B$$
$$= \overline{B} + \overline{C} + \overline{A}\,B$$

D. $\overline{B} + \overline{C} + \overline{A}\,B$

3. Data Structures

The binary search tree for TOKYOOLYMPICS is as follows:



E. I O S

4. Data Structures

The queue is constructed using FIFO as follows:

A, AQ, AQU, QU, QUA, QUAR, UAR, AR, ARI, RI, RIU, RIUS

C. R

5. FSAs and Regular Expressions

[ACSL]*[^aeiou]*21.2

a.	Ab212	Fails - needs a character between 1 and 2
b.	ACSL21	Fails - missing a character and 2 at the end
c.	ACSL442122	Valid
d.	cmptr212	Fails - needs a character between 1 and 2
e.	A202122	Valid
f.	pgrmmng21	Fails - missing a character and 2 at the end

C. c, e

6. Boolean Algebra

$$\overline{A\,\overline{B}} + \overline{\overline{A}\,B\,\overline{C}} + \overline{\overline{B}\,C} = (\overline{A} + B) + (A + \overline{B} + C) + (B + \overline{C})$$
$$= (A + \overline{A}) + (B + \overline{B}) + (C + \overline{C})$$
$$= 1$$

B. 1

7. Boolean Algebra

$A\overline{B} + \overline{A}(B + \overline{C}) = A\overline{B} + \overline{A}B + \overline{A}\,\overline{C}$

If $A = 1$, then $\overline{B} + 0 + 0 = 1$ So $\overline{B} = 1$ (1, 0, *) makes it TRUE

If $A = 0$, then $0 + B + \overline{C} = 1$

If $B = 1$, then $C =$ * (0, 1, *) makes it TRUE

If $B = 0$, then $0 + 0 + \overline{C} = 1$ (0, 0, 0) makes it TRUE

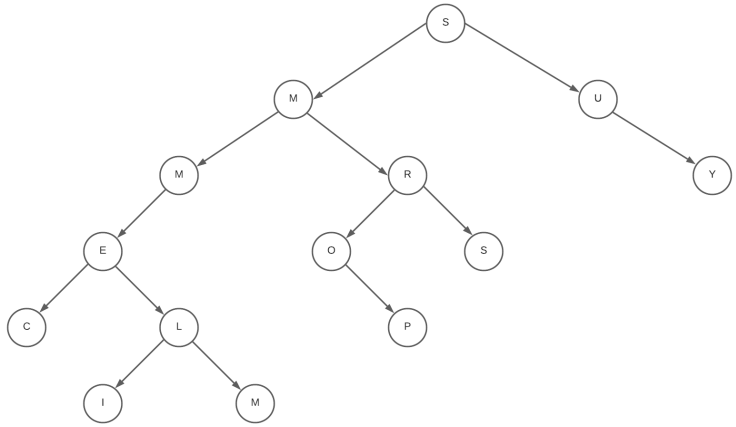
Therefore, 5 ordered triples make it TRUE.

C. 5

8. Data Structures

The binary search tree for SUMMEROLYMPICS is:

C. 2



Only 2 nodes have only a right child: O and U

9. Data Structures

The stack is constructed using LIFO as follows:

C, CA, CAN, CA, C, CI, C, CS, CSM, CS, C, CA,
CAJ, CA, CAO, CA, CAR, CA

The next item popped would be: A

E. A

10. FSAs and Regular Expressions

1-321-[^05][0-9][^5]-[12][3-6][^7-9]8-[^0][2-6]

a. 1-321-123-2348-11 Fails at 5th 1
b. 1-321-155-2368-96 Fails at 2nd 5
c. 1-321-444-2448-44 Valid
d. 1-321-927-2408-84 Valid
e. 1-321-559-2328-13 Fails at 1st 5
f. 1-321-306-1366-62 Fails at 3rd 6

D. c, d